

The Grounded Research Object

Metric-informed design of a machine-native scientific record whose unit computes its own credibility

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ABSTRACT. Research assessment runs on citation-derived indicators despite two decades of documented dysfunction: citations reward citable outputs and are structurally blind to negative results, replications, refutations, and reuse. I ask whether indicators derived from the *structure* of a machine-readable research record can measure — and let funders reward — what citations cannot. To test this I drafted 64 candidate contribution indicators over a structured corpus — an existing agent-native record format (ARA; Liu et al., 2026) adopted as the departure point for this work — and selected them through blind adversarial tournaments. The central result is negative and clarifying:

METRICS COMPUTED OVER A RECORD'S OWN STRUCTURE MEASURE THE FIDELITY OF THE RECORD, NOT THE QUALITY OF THE SCIENCE; genuine signal appeared only at the claim level, anchored to external ground truth. Auditing every blocked indicator against the data shape that blocked it shows most failures are format problems, not feasibility problems — the record does not lack the knowledge, it lacks the shape. From that audit I derive the Grounded Research Object (GRO): a record format that sorts every quality signal into three rigor tiers — deterministic, reliable-anchored, and reproducible-judged — kept physically separate so no self-certified number is mistaken for a checked one, and rolled into a funder-facing credibility dossier that never collapses to a single number. GRO is a specification, adversarially stress-tested but empirically unvalidated; I state its limits plainly and argue the two binding constraints — validation and adoption — are not protocol problems.

1. Introduction

The paper is the unit of scientific output, and the citation is its score. Once that holds, Goodhart's law does the rest: when a measure becomes a target it ceases to be a good measure. Careers, grants, and tenure denominated in citations push researchers toward citable work — reviews, positive results, broad confident narratives — and away from the labor everyone agrees is undervalued: negative results, replications, refutations, shared data, methods reused downstream. The reform literature (DORA; the Leiden Manifesto; *The Metric Tide*; CoARA) is mature but largely *subtractive* — use fewer numbers, more narrative — with few constructive proposals for indicators that measure the neglected contributions, and with Goodhart usually asserted against any new metric rather than tested.

Meanwhile the record itself is changing. AI agents increasingly read, reproduce, and extend research, and machine-readable structured records — typed claims, evidence links, exploration traces — are emerging as a publication layer. For the first time there are artifact surfaces over which non-citation indicators could actually be computed. This paper reports what happens when you try, and what the attempt reveals about the record we would need. The governing thesis is blunt: **reward what citations punish**, and defend against gaming not with one clever number but with a diversity of independent signals and a selection process in which no metric is graded by its own author.

2. The metrics program

I organised the work around seven directions of good-science signal: (1) reproducibility and specification completeness; (2) claim and evidence integrity; (3) process transparency and *negative knowledge* — dead-end density, failure-knowledge preservation, negative-result share; (4) novelty and dependency structure, including a corrective-science score weighting refutations above imports; (5) a cross-library *claim graph* whose edges are evidential (corroboration, contradiction, replication depth, reuse) rather than social — a direct alternative to the citation network; (6) real-world grounding against trial registries, chemical databases, and datasets; and (7) representation quality as a trust weight.

These became 64 concrete candidate indicators drafted across every surface a structured record exposes, pruned and merged to 23 survivors and a top ten. Selection ran as two blind, adversarial tournaments so a metric could not be graded by whoever proposed it. In round one, for each of eleven artifact surfaces, four clean-room proposers (no sight of any existing metric) proposed indicators; a meta-science judge picked the best two; both were revised against critique; one won. In round two, each top-ten survivor ran a four-to-two-to-one tournament over three refinement cycles. Proposers were drawn from two independent model families (GPT-5.5/codex and Claude Sonnet 5) and the meta-science judge was a separate model (Claude Fable), so no metric was graded by the model that proposed it. The tournaments were run over a real corpus — roughly 140 Alzheimer's-domain papers, about 60 compiled into the ARA format (Liu et al., 2026) — so the indicators were not merely designed but *run*, and measured.

3. Result: structure measures fidelity, not quality

The headline is negative, and it is the most useful output of the exercise:

Metrics computed over a record's own structure measure the fidelity of the record, not the quality of the science.

A well-compiled record of bad science and a well-compiled record of good science were indistinguishable to every structural metric in the pool. Zero of six paper-level rankers survived the validity check. Genuine signal appeared only at the claim level, and only when it reached *outside* the record: claims checked against registered clinical-trial endpoints, priority claims checked against prior literature, and one compiler-introduced polarity inversion caught. A direct measurement of extractive fidelity against full text ran at roughly 92% faithful, with the failures concentrated in dense multi-cohort comparison papers where a transposed table flips a result.

The lesson is not that structural metrics are impossible. It is sharper: the signal worth rewarding lives outside the record's own walls, and the record's *shape* determines whether you can reach it.

4. The affordance gap

For every metric I wanted but could not build, I asked why. Almost never because the information was missing — because it was stored in a shape only a language model could re-extract: a number hand-retyped as prose in four places instead of one typed value, citations as author-year strings instead of resolvable identifiers, honest absence indistinguishable from lazy omission. The record did not lack the knowledge; it lacked the shape. Every blocked metric sorts into one of three classes, and each tells the format what to do.

Class	Why the metric is blocked	What the format must do
Format-recoverable	the fact is present but as prose an LLM must re-read	emit it typed → the metric becomes a deterministic join
Anchor-dependent	the fact points outside the record and nothing resolves the pointer	guarantee resolvable external IDs so registries and indices can be joined
Irreducibly semantic	no format change makes it computable	a calibrated judge, forever — the format only feeds it cleaner, pinned inputs

The first two classes are format problems masquerading as feasibility problems. Only the third is a true ceiling. This taxonomy is the bridge from a metrics result to a substrate design.

5. The Grounded Research Object

GRO gives the record a shape. Every load-bearing fact exists once as a canonical typed record, addressed by an identifier; prose binds back to it. The narrative paper survives as a *compiled view*, not the source of truth. Everything a record must carry is sorted into three rigor tiers, kept in physically separate files so a self-certified number can never be dressed as a checked one.

Tier	What it holds	How far it can be trusted
DETERMINISTIC	typed quantities, claim logical form, typed cross-layer graph, genre manifest	exact and reproducible — a structural join
RELIABLE-ANCHORED	references, registrations, accessions, datasets	reliable given a pinned resolver; failures quarantined, not faked
REPRODUCIBLE-JUDGED	novelty, significance, entailment quality, assumption realism	a calibrated judge, forever; the format only feeds it cleaner inputs

A decisive structural invariant separates two write-surfaces: the compiler writes only the skeleton facts it can ground, while independent resolver, extraction, and judge passes write everything that *certifies* those facts, into append-only files a validator forbids the compiler from touching. The tiers roll up into a funder-facing credibility dossier that never collapses to one number — a program officer can gate on the deterministic floor, rank on the judged tier, read the rest as risk, and audit any score back to its source *and its rigor class*. Honest absence — `not_specified`, `not_applicable`, `unresolved` — is a first-class value scored equal to an honest answer, so a typed field never creates pressure to fabricate.

The design was not hand-authored. Each of twelve affordance gaps ran its own four-proposer design tournament; the twelve winning designs were merged into one specification and put through an adversarial red-team pass that forced every over-claim down to a stated limitation. GRO is a clean-sheet design informed by, but not inheriting from, the format it critiques.

6. Where existing efforts fall short

Three current efforts each measure a different axis, none calibrated against the others, and none covering the reward-what-citations-punish set. **Citations** measure social attention. A structural **rigor seal** (the closest kind of automated review) measures internal *integrity* — coherence, falsifiability, grounding — as a single grade, but never asks whether the work is novel or matters, and is calibrated to nothing external. External benchmarks such as the Metascience Novelty Indicators Challenge measure *novelty*, calibrated to expert ratings, but only novelty, and they read whole pa-

pers rather than a structured record. Integrity, novelty, and fidelity are three orthogonal axes; the contributions citations punish fall between them. Closing that gap is a substrate problem first — most of the ideal indicator set cannot be computed until the record carries the right shapes.

7. Limitations

GRO is a specification, adversarially hardened but **empirically unvalidated**. The honest limits are as important as the design, and several are drawn directly from the two research programs this work feeds.

7.1 It has not been shown to discriminate

The open experiment — the one that would tell us whether any of this is real — has not been run: take the affordance-derived metrics, run them over existing literature (starting from the testbed corpus, adding a contrasting second domain), and validate against external ground truth such as trial-registry concordance, retraction and correction records, and prior-literature novelty. Until then, "GRO makes the metric computable" is not "the metric separates good science from bad." The negative result of §3 is a warning that this must be shown, not assumed. Running that discrimination test — extending these metrics over a labelled corpus and calibrating them against funder judgements of what to reward — is the subject of the author's doctoral programme at SPRU (University of Sussex). **Update (2026-07): a first discrimination test has now been run**, on the breakthrough axis, over 66 recent and 72 historical (2004–2010) Alzheimer's papers (see repository `experiment/breakthrough/`). The result is sobering and is reported in full: the one metric that carries signal is the L8 contribution-typing (`max(peak, cwmean)`). Against a same-model LLM expert panel it reached $\rho \approx 0.58$ (held-out); against an independent model family only $\rho \approx 0.34$ (roughly a third was shared-method bias — the lean flips when the metric is built by the other model, Steiger $p=0.003$); and against a *real-world, LLM-free ground truth* — mature citation-disruption of the historical papers over the following 15–20 years — it showed **no reliable predictive power** (ρ near zero, sign-unstable). The prior-art `overlap/sota_anchor` axis, the delta ledger, and genre added no transferable signal. So on the evidence so far the metric flags *LLM-perceived contribution depth*, which is not shown to be *field-resaping impact*. This sharpens rather than removes §7.1: the discrimination test is now begun, its first verdict is largely negative, and the honest next step (a full-text, multi-domain historical corpus, calibrated against real outcomes) is the subject of the author's doctoral programme at SPRU (University of Sussex).

7.2 Importance is judged, not computed — and does not scale

Standardisation buys *fidelity*-measurability cheaply and deterministically. It does not buy *importance*-measurability: novelty, validity, and contribution significance remain in the judged tier, needing a calibrated judge and human-labelled ground truth. That tier does not scale to a full corpus — deterministic and anchored signals can run at population scale, but calibrated judgment runs only on a triage set plus an audit sample. The corpus-scale promise is therefore honestly "deterministic and anchored credibility at scale; calibrated-judged credibility at triage-and-sample scale," which means the format does not, on its own, surface the un-shortlisted gem.

7.3 A record that computes its own credibility relocates trust; it does not eliminate it

The anti-fabrication guarantees rest on an independence between automated checkers that is, throughout, *asserted rather than measured*: genuinely decorrelated checking is earned at exactly one non-LLM point in the design; everywhere else "independent" means "a different model," and

two models can share a blind spot. So credibility is moved from trusting the author's prose to trusting a calibration set and an assumed independence — real progress, because the trust is now auditable, but not the free lunch the phrase suggests. Calibration sets and probe sets are themselves institutional trust anchors that can be captured, and cross-discipline comparison remains a modelling choice, withheld rather than faked where a discipline cannot be resolved.

7.4 Adoption is the wall, and it is owned by incentives

Every prior attempt at a better medium — nanopublications, research objects, and my own earlier work — failed not on architecture but on incentives: no reason for a researcher to assemble structured objects when grants and jobs are denominated in papers. A better specification does not move that wall. What has changed is the cost side: agents now sit inside the research workflow and the trace they generate is the raw material, so capture can be largely automated — necessary, nowhere near sufficient. And the record is extensible by design: agent-native substrates such as ARA (Liu et al., 2026) and Claude Science take new data shapes through adapters and plugins scoped to the target metric, so the format can grow to carry a chosen indicator's inputs without a rewrite. The binding questions — which field's researchers are the real early adopters, and whether funder co-design can shift incentives at meaningful scale — are not protocol problems, and this work does not solve them. They are, deliberately, the subject of the applied research programme this specification supports: the empirical Goodhart test of these indicators under adversarial optimisation, and the study of what evaluation arrangements funders and evaluators would regard as legitimate.

8. Next steps

Two moves, both closing the gap between "designed" and "shown to work." First, **run the discrimination experiment** (§7.1): compile existing literature into the format and validate the metrics against external ground truth, testing at the same time whether the three affordance classes hold under adversarial pressure. Second, **build the metrics with funders**. Funders are where measurable becomes rewarded; the metrics over the record should be shaped by what funders actually want to reward, and the affordance model tells them what the record must carry for a chosen indicator to be valid rather than gameable. Concretely: name the undervalued contributions to reward; supply the human-labelled ground truth (as the Novelty Challenge did for one axis); and fund the shapes, resolvers, and calibration sets as shared public goods. Non-traditional funders are already piloting new publishing practices as funding conditions; the larger innovation agencies could sharpen their own metric efforts with an affordance-aware view of the record.

9. Conclusion

Chasing good metrics far enough hands you the specification for the record itself. The metrics are the entry point — the lever, because nothing changes until what counts changes, and the diagnostic, because trying to compute them reveals what the substrate must be. GRO is that substrate, designed backwards from a metrics failure rather than forwards from taste, and honest about the two things still missing: proof that the metrics discriminate, and an institution willing to make the good ones count. Neither is a protocol problem, which is precisely why the work now turns to experiment and to funders.

Appendix. Method and provenance

The metric indicators were selected by two blind adversarial tournaments (round one per artifact surface, ~88 subagents; round two per metric, four-to-two-to-one over three refine cycles). The GRO specification was produced by twelve per-gap design tournaments (four independent proposers each — two on one model family, two on another — a critical meta-science judge selecting and critiquing the best two, a refine round) whose twelve winning designs were merged and then subjected to an adversarial critique pass that forced every over-claim to an explicit limitation or a design change; design-tournament run identifiers `wf_f0bc615b-a88` and tail `wf_c4cbff37-887`. All artifacts — the seven directions, the 64→23→10 indicator ledger, the negative-result report, the affordance-gap taxonomy, the full GRO specification, both tournaments with their winning proposals and judgements, and the compute code including the cross-library claim graph — are collected in the accompanying repository (github.com/carlaost/gro). This document is a digest of a longer compiled reference; where it summarises, the repository carries the full text.

This work adopts the ARA format (Liu et al., 2026) as the substrate it critiques and stress-tests; GRO is the substrate-design output of that critique, not a re-implementation. Positioning, evidence pointers, and the honest-limitations detail are documented in the repository and in the author's research program map.